DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 11

BigQuery beyond the queries, and predicting a number

From a salary line to medical charges: your first models

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Sun 18 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Place machine learning inside AI

02 Name features, label, training and inference

03 Train, read and use a LINEAR_REG model

04 Hold rows aside; score with MAE and R^2

05 Beat the average guess; keep the simpler model

EVERYONE DO

Step 1.1

Warm-up

Questions on D10’s window functions.

Suggested time · 10 minutes

PART 01

Recap: BigQuery beyond the queries

D10’s last part, in eight minutes. The labs are yours to explore.

INSTRUCTOR DO

Step 2.1

Recap: BigQuery beyond the queries

Storage and compute, files read in place, partitions, and the map.

Suggested time · 8 minutes

labs/D11/01-Ins-BigQuery-Map/README.md

STORAGE AND COMPUTE

Two separate parts of a cloud warehouse

Storage

tables and files; cheap; always there

Compute

starts when you run a query; billed by bytes read

NATIVE OR EXTERNAL?

Load the rows, or read them in place

	Native table	External table
Rows are	in BigQuery’s storage	in a file in Cloud Storage
Storage size	shown in Details	none
File changes	upload again	the next query reads it
Speed	fast	slower: read every time

PARTITION PRUNING

A query for April reads one month

Picture nakheel.orders split by month: 24 partitions.

2024-09	2024-10	2024-11	2024-12	2025-01	2025-02	2025-03	2025-04	2025-05	2025-06	2025-07	2025-08
2025-09	2025-10	2025-11	2025-12	2026-01	2026-02	2026-03	**2026-04**	2026-05	2026-06	2026-07	2026-08

```
WHERE order_date >= '2026-04-01' AND order_date < '2026-05-01'
```

BigQuery reads one partition: 144 of the 3,000 orders.

What you have used, and what needs billing

Area	In BigQuery	In the sandbox
Storage, compute	tables; SQL queries	yes: 10 GiB, 1 TiB a month, 60 days
Load, read in place	uploads; external tables	yes; your own bucket: billing
Save work	views; CREATE TABLE AS	yes
Change rows	INSERT, UPDATE; schedules	no
Read less	partitions; clusters	yes, with the 60-day catch

Beyond SQL

Area	In BigQuery	In the sandbox
Machine learning	built-in models; trees; AutoML	built-in: yes
Gemini models	called from SQL	no: needs billing
AI help in the editor	Gemini: write, explain SQL	not used here
Python	notebooks, DataFrames	not used here
Governance	roles, lineage, quality scans	roles yes; the rest no

AFTER CLASS

Three D10 labs to try at home

Lab 09

an external table on Google’s public file

Lab 10

what the sandbox shows, and what needs billing

Lab 14

partitions and clusters: read less

labs/D11/01-Ins-BigQuery-Map/README.md

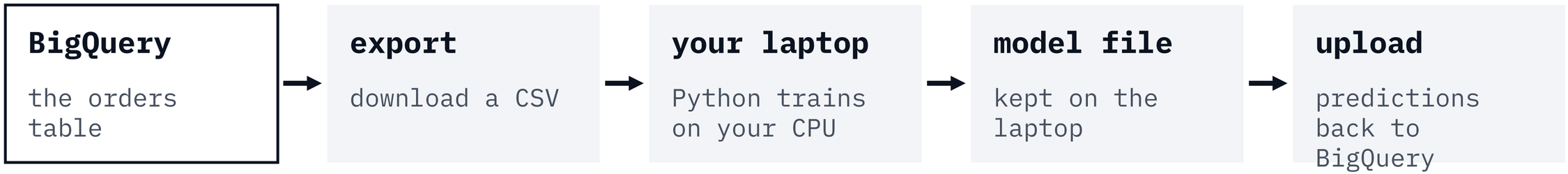
DEFINITION

BigQuery ML builds and uses models with SQL, inside the warehouse, where the data already is.

WHY TRAIN WHERE THE DATA LIVES

Move the data to the model, or the model to the data

PYTHON ON YOUR LAPTOP



BIGQUERY ML



With BigQuery ML the data does not move: BigQuery's engine trains the model, and a prediction is one query away.

Until now you have used SQL to understand what happened. Today you use SQL to learn from history and estimate what may happen next.

PART 02

A first model, before the theory

Thirty past hires: upload them, train a model, read it, ask it about a new candidate.

INSTRUCTOR DO

Step 3.1

Today’s question

Thirty past hires, and one new candidate.

Suggested time · 4 minutes

What salary should we expect for a candidate with 5 years of experience?

FIVE OF 30 PAST HIRES · SALARY DATA

The candidate is not in the table

	years of experience	salary (USD)
a past hire	1.1	39,343
a past hire	3.2	54,445
a past hire	5.1	66,029
a past hire	10.5	121,872
the new candidate	5.0	**?**

One row is one past hire: what they brought, and what they were paid.

EVERYONE DO

Step 3.2

Upload and look

Make the ml dataset, upload the file, look at all 30 rows.

Suggested time · 10 minutes

labs/D11/02-Evr-First-Model/README.md

A home for this week’s models

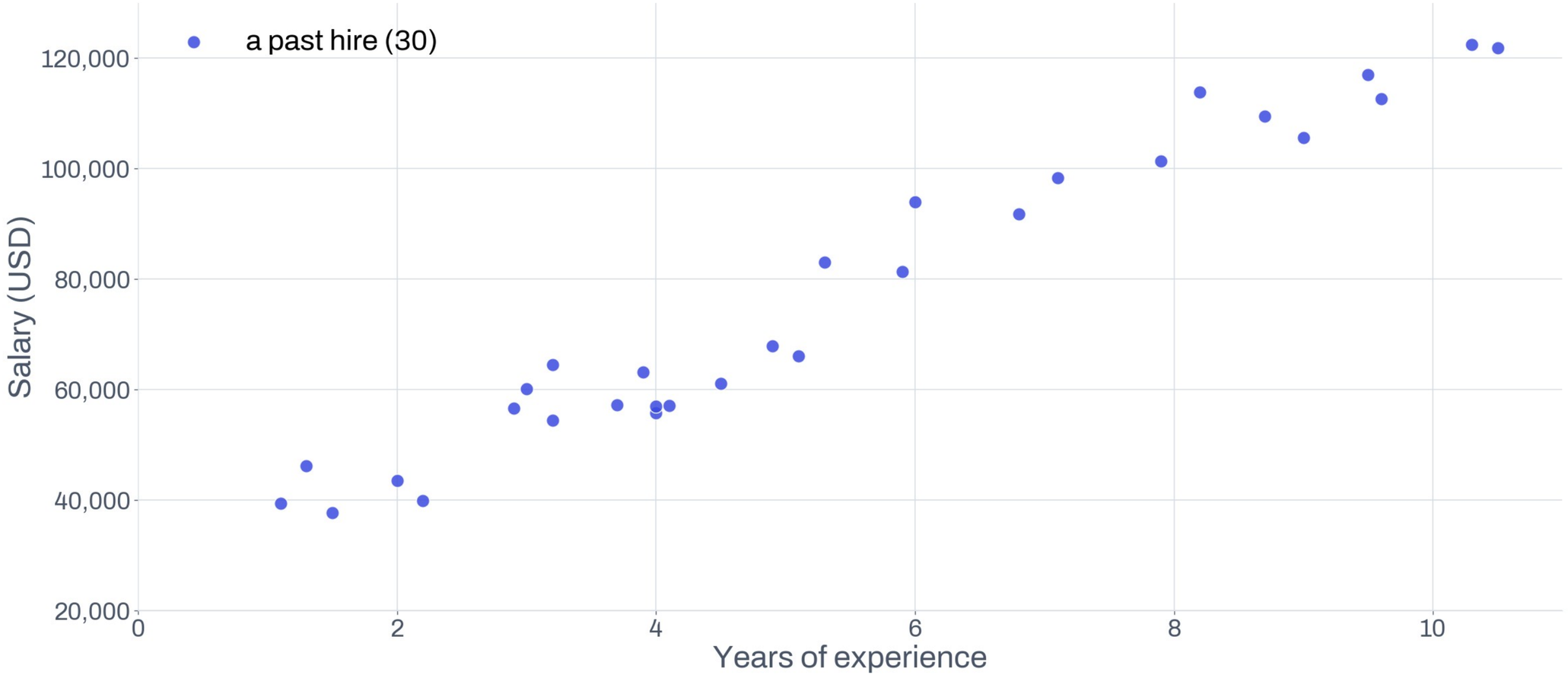
CREATE SCHEMA makes a dataset; models live in datasets, like tables.

```
CREATE SCHEMA IF NOT EXISTS ml OPTIONS (location = 'US');
```

Then upload salary_data.csv as ml.salary_data: Create table, Upload, CSV, Auto detect.

BLOCK 2 · ALL 30 PAST HIRES

Salary rises with experience



Salary data, 30 rows, US dollars. 1.1 to 10.5 years; about 37,700 to 122,400 dollars.

EVERYONE DO

Step 3.3

Train it, read it, ask it

Training takes about a minute.

Suggested time · 12 minutes

labs/D11/02-Evr-First-Model/README.md

Which method

`model_type = 'LINEAR_REG'`: linear regression predicts a number.

```
CREATE OR REPLACE MODEL ml.salary_model
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['salary']
) AS
SELECT years_experience, salary
FROM ml.salary_data;
```

Which column is the answer

input_label_cols: the label. Every other selected column is a feature.

```
CREATE OR REPLACE MODEL ml.salary_model
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['salary']
) AS
SELECT years_experience, salary
FROM ml.salary_data;
```

A model is a set of numbers

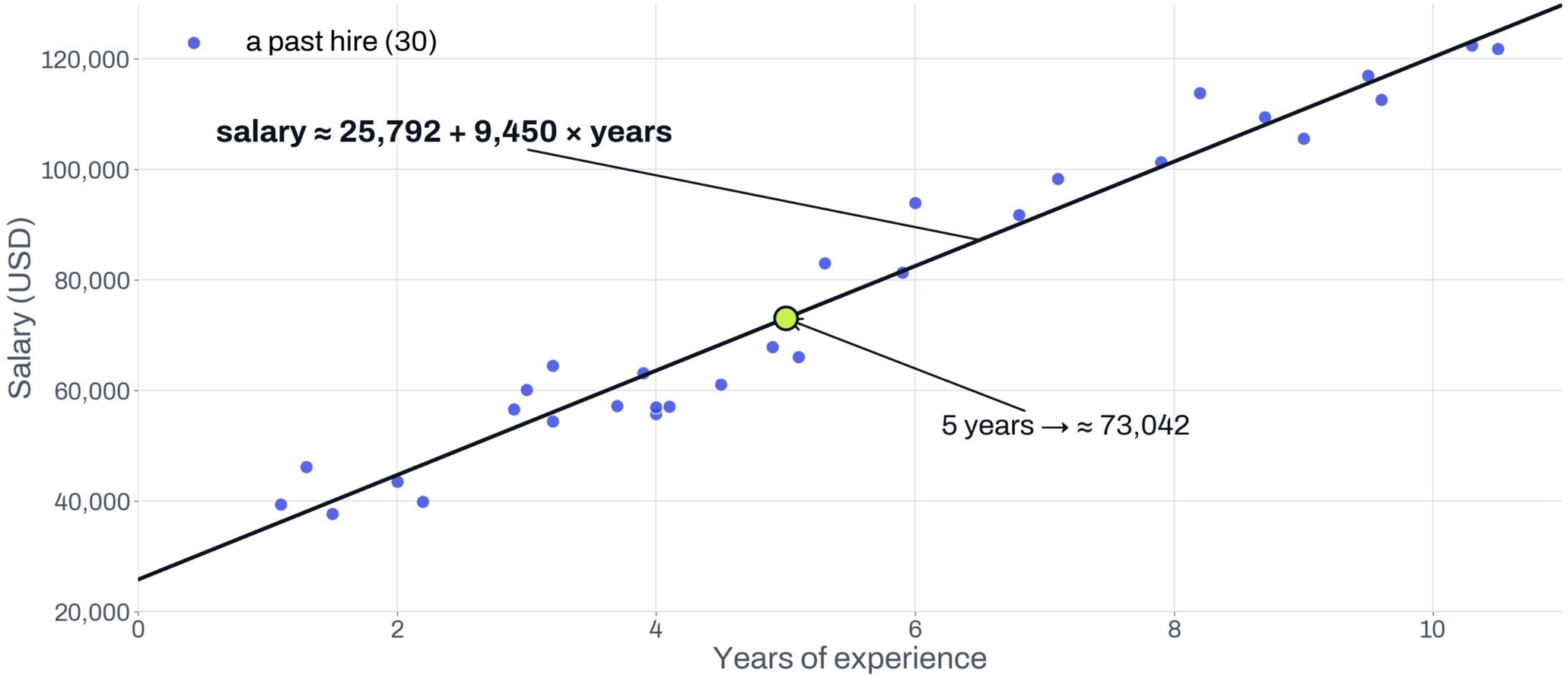
ML.WEIGHTS shows the numbers training chose.

```
SELECT processed_input, ROUND(weight, 1) AS weight
FROM ML.WEIGHTS(MODEL ml.salary_model);
```

→ 2 rows · intercept ≈ 25,792 · years ≈ 9,450

THE MODEL IS A LINE

Start at 25,792; add 9,450 a year



Training chose the line whose squared misses on the 30 hires add up to the least.

Ask it about the new candidate

ML.PREDICT adds predicted_salary to the rows you give it.

```
SELECT years_experience,
       ROUND(predicted_salary) AS predicted_salary
FROM ML.PREDICT(MODEL ml.salary_model,
  (SELECT 5.0 AS years_experience));
```

→ 1 row · predicted_salary ≈ 73,042

Check it by hand: 25,792 + 9,450 × 5 = 73,042.

THE QUESTION FOR THE NEXT HOUR

Is this model any good? How would we know?

PART 03

What just happened?

Where machine learning sits, how a model is learned, and the words for it.

INSTRUCTOR DO

Step 4.1

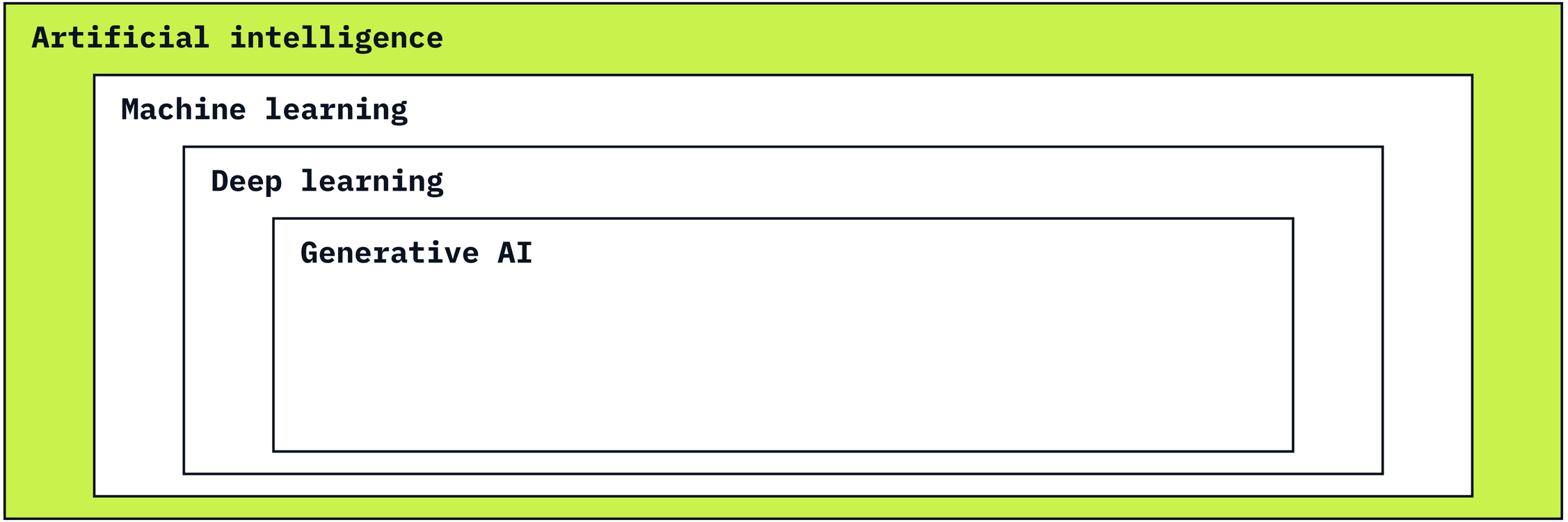
Where machine learning sits

Four rings, one at a time, and a definition.

Suggested time · 6 minutes

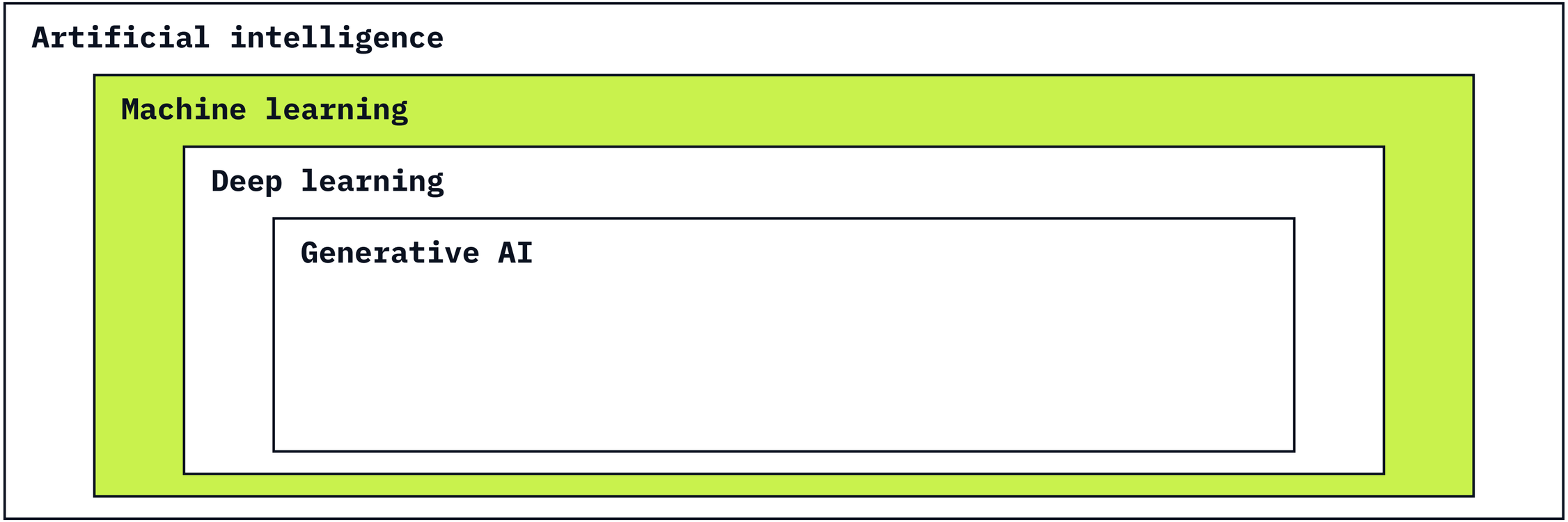
AI is the umbrella; each ring sits inside the one around it

Artificial intelligence: the umbrella term for any system doing a task we call intelligent.



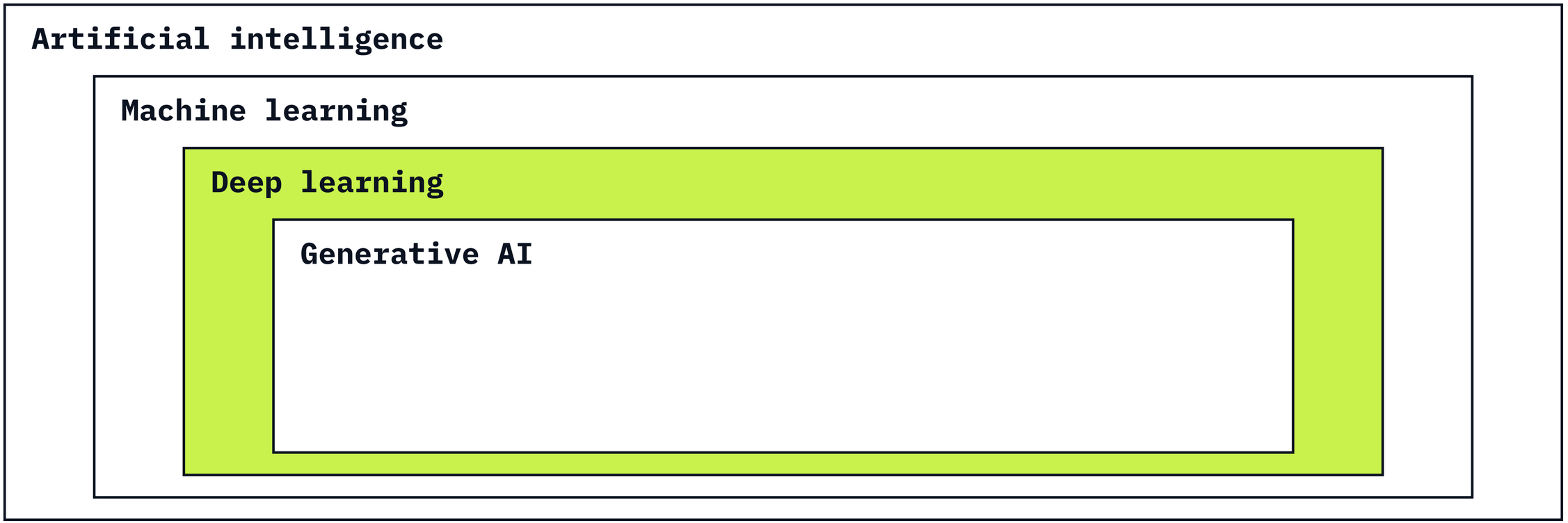
AI is the umbrella; each ring sits inside the one around it

Machine learning: rules learned from examples. Your model today: learned from 30 past hires.



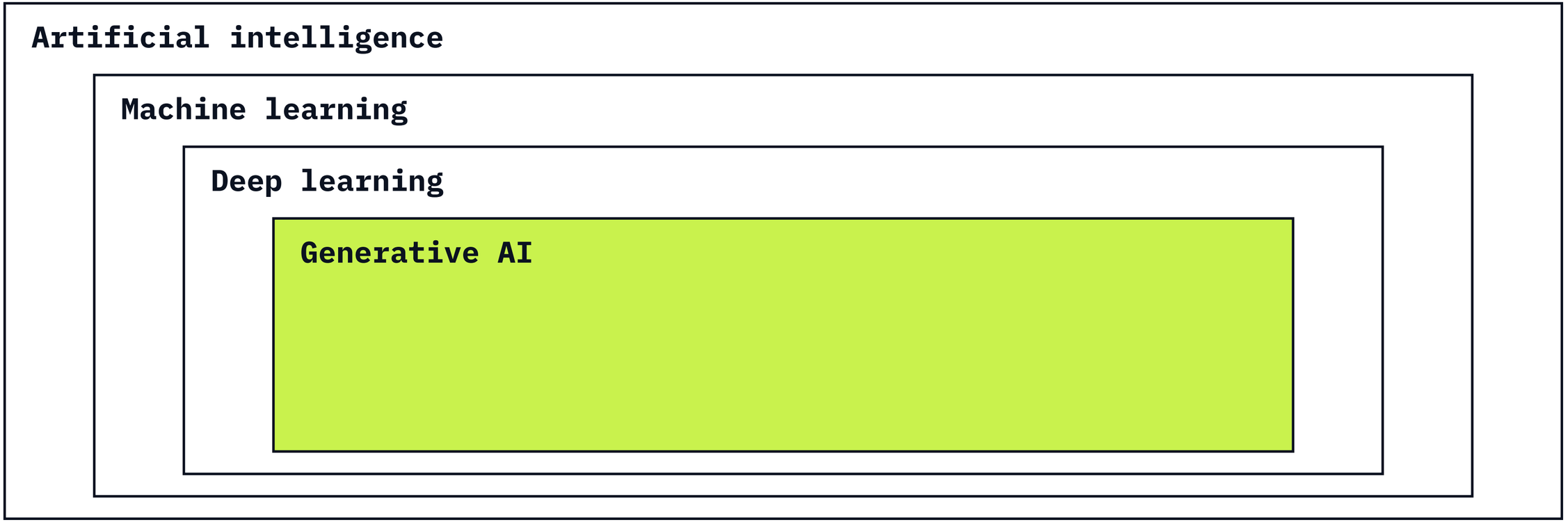
AI is the umbrella; each ring sits inside the one around it

Deep learning: many-layered neural networks that learn their own features.



AI is the umbrella; each ring sits inside the one around it

Generative AI: models that produce new text (large language models) or images; built mostly on deep learning.



DEFINITION

Machine learning is a program whose rules an algorithm finds from recorded examples.

Ordinary software runs rules a person wrote. A model is the rule the algorithm wrote. In supervised learning, the examples carry the answer.

INSTRUCTOR DO

Step 4.2

Write the rule, or learn it

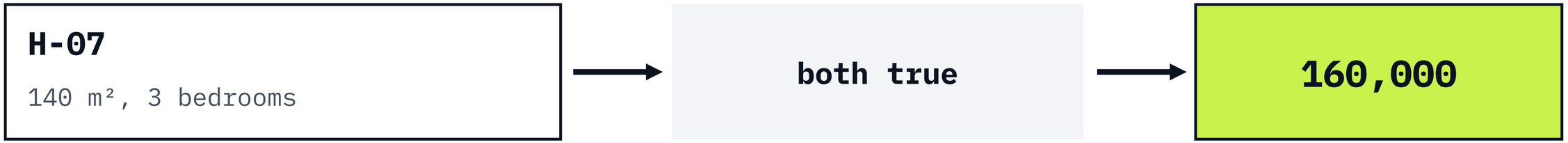
A small, invented house example: a rule, then past sales.

Suggested time · 8 minutes

ILLUSTRATIVE · INVENTED HOUSES AND PRICES

A person writes a price rule

If a house has at least 120 m² and at least three bedrooms, estimate **160,000 JOD**; otherwise **100,000 JOD**.



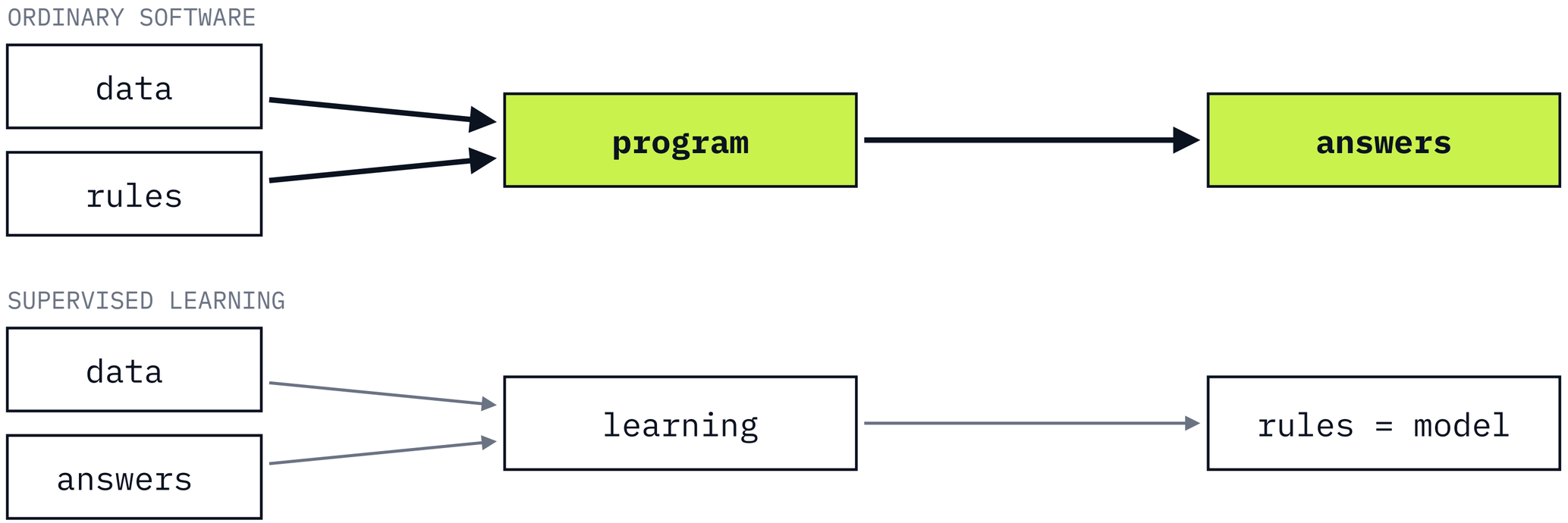
A HOUSE TO PRICE (REFERENCE H-07)

It ignores the street, the age, the view: a decision put into a rule.

TWO WAYS TO GET A RULE

Write the rules, or learn them from answers

Ordinary software: data and rules go in, answers come out. You wrote every rule.



ILLUSTRATIVE · INVENTED HOUSES AND PRICES

Or learn from houses that have sold

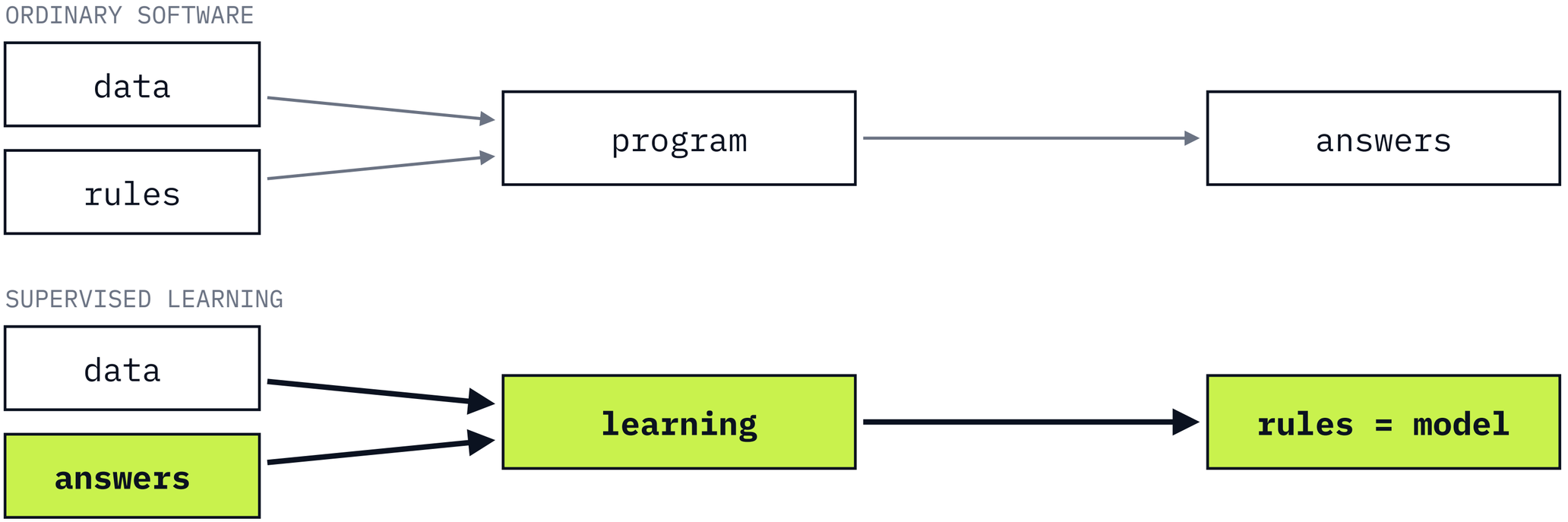
house	floor area m²	bedrooms	sale price JOD
H-01	95	2	82,000
H-02	120	3	118,000
H-03	150	3	141,000
H-04	180	4	170,000
H-05	110	2	99,000
H-06	165	4	152,000

A training procedure fits a model to these sales, instead of a person choosing every threshold and price.

TWO WAYS TO GET A RULE

Write the rules, or learn them from answers

Supervised learning: data and recorded answers go in, the rules come out.



INSTRUCTOR DO

Step 4.3

One table, five roles

The same houses, one role lit at a time; then the words.

Suggested time · 7 minutes

ILLUSTRATIVE · INVENTED HOUSES AND PRICES · 1 OF 4

One row: one completed sale

house	floor area m²	bedrooms	sale price JOD
H-01	95	2	82,000
H-02	120	3	118,000
H-03	**150**	**3**	**141,000**
H-04	180	4	170,000
H-05	110	2	99,000
H-06	165	4	152,000

An example (also: observation, sample). From now on: example.

Floor area and bedrooms: the features

house	floor area m²	bedrooms	sale price JOD
H-01	**95**	**2**	82,000
H-02	**120**	**3**	118,000
H-03	**150**	**3**	141,000
H-04	**180**	**4**	170,000
H-05	**110**	**2**	99,000
H-06	**165**	**4**	152,000

Features: the inputs the model reads.

Sale price: the label

house	floor area m²	bedrooms	sale price JOD
H-01	95	2	**82,000**
H-02	120	3	**118,000**
H-03	150	3	**141,000**
H-04	180	4	**170,000**
H-05	110	2	**99,000**
H-06	165	4	**152,000**

The label (or target): the value to predict.

The house reference: an identifier

house	floor area m²	bedrooms	sale price JOD
H-01	95	2	82,000
H-02	120	3	118,000
H-03	150	3	141,000
H-04	180	4	170,000
H-05	110	2	99,000
H-06	165	4	152,000

An identifier tracks the row. It is not an input: a house number says nothing about the price.

SIX WORDS FOR THE WEEK

Each word, and where you saw it

Word	Meaning	In the house table
Example	one row: one completed sale	H-03
Feature	an input the model reads	area, bedrooms
Label	the value to predict	sale price
Identifier	tracks the row; not an input	house
Model	the learned relationship: fitted numbers	out of training
Prediction	the model's estimate; can differ from the actual	out of inference

INSTRUCTOR DO

Step 4.4

Kinds of learning

Four kinds, one at a time; then: a number or a category?

Suggested time · 5 minutes

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Supervised: a salary, medical charges (today), will an employee leave (D12).

Kind	The rows hold	Typical tasks
Supervised	**features and the label**	**regression, classification**
Unsupervised	features only	clustering, fewer columns (PCA), anomalies
Semi-supervised	a few labeled rows, many unlabeled	label a little, learn from all
Reinforcement	actions and rewards	an agent learning by trial

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Unsupervised: customer segments, PCA, unusual transactions. D13.

Kind	The rows hold	Typical tasks
Supervised	features and the label	regression, classification
Unsupervised	**features only**	**clustering, fewer columns (PCA), anomalies**
Semi-supervised	a few labeled rows, many unlabeled	label a little, learn from all
Reinforcement	actions and rewards	an agent learning by trial

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Semi-supervised: analysts confirm a few hundred fraud cases; millions of payments stay unlabeled.

Kind	The rows hold	Typical tasks
Supervised	features and the label	regression, classification
Unsupervised	features only	clustering, fewer columns (PCA), anomalies
Semi-supervised	**a few labeled rows, many unlabeled**	**label a little, learn from all**
Reinforcement	actions and rewards	an agent learning by trial

KINDS OF MACHINE LEARNING

Four kinds, told apart by what the rows hold

Reinforcement: pricing or robot control by trial and reward. Named here, not taught.

Kind	The rows hold	Typical tasks
Supervised	features and the label	regression, classification
Unsupervised	features only	clustering, fewer columns (PCA), anomalies
Semi-supervised	a few labeled rows, many unlabeled	label a little, learn from all
Reinforcement	**actions and rewards**	**an agent learning by trial**

DEFINITION

Supervised learning learns from rows that carry the answer, then predicts it for new rows.

Features (X): columns the model reads. Label (y): the column it predicts.

TWO KINDS OF SUPERVISED LEARNING

Regression or classification?

Ask what the label holds: a number, or a group?

REGRESSION

Label: a number

Predicts how much

House price

CLASSIFICATION

Label: a category

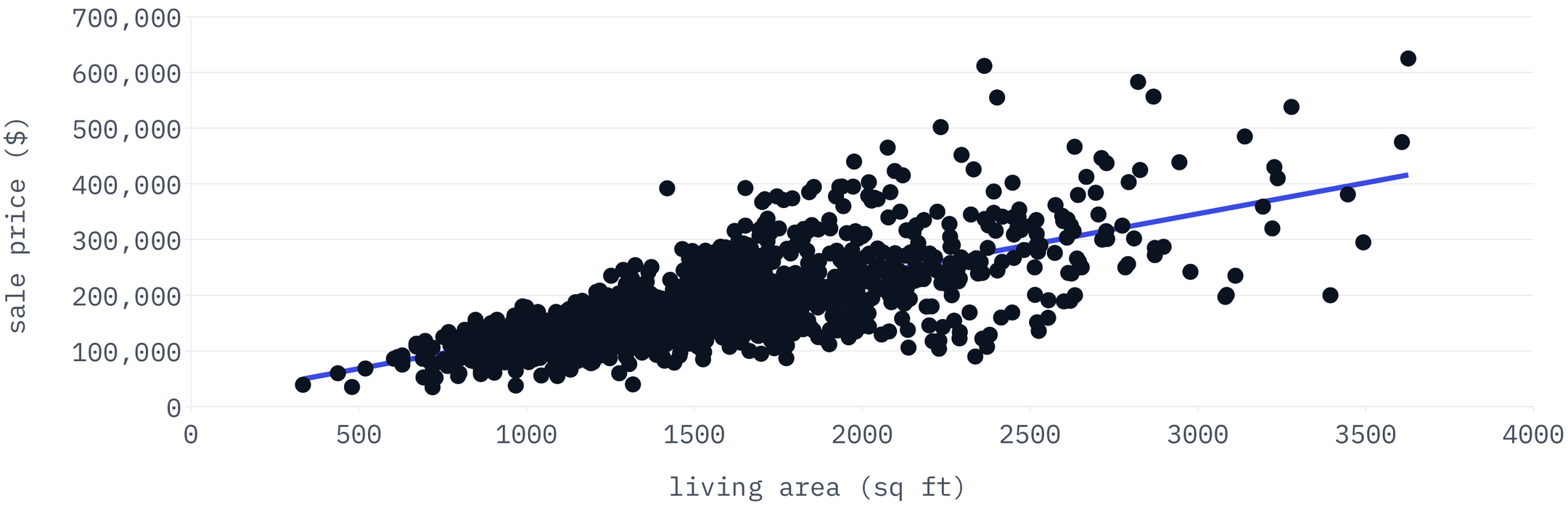
Predicts which group

Churn yes or no

REGRESSION · AMES HOUSE SALES

Regression estimates a number

Here a straight line fits the sales: about $111 more per square foot.



EVERYONE DO

Step 4.5

Which kind of problem?

Four requests. For each: one row, and the kind of learning, or a rule.

Suggested time · 5 minutes

FOUR REQUESTS

Which kind of problem?

#	Request
1	Finance: how much will each customer spend in the next six months?
2	Marketing: which customers will order again in the next six months?
3	Marketing: which customers behave alike, so we can treat them as groups?
4	Operations: charge 3 JOD delivery on orders under 20 JOD.

THE ANSWERS

One row first, then the kind

#	One row	Kind
1	a customer	**supervised: regression, a number (today)**
2	a customer	supervised: classification, yes or no (D12)
3	a customer	unsupervised: clustering, no label (D13)
4	an order	no ML: the rule is known exactly; a CASE

INSTRUCTOR DO

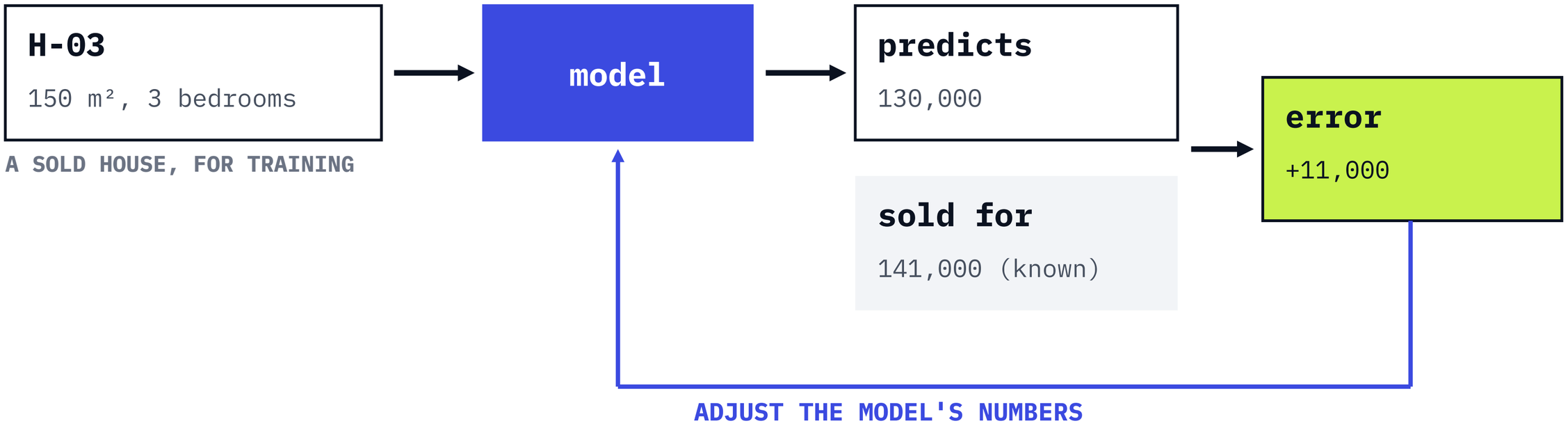
Step 4.6

Training, inference, and checking

Three pictures with the same houses: learn, predict, check.

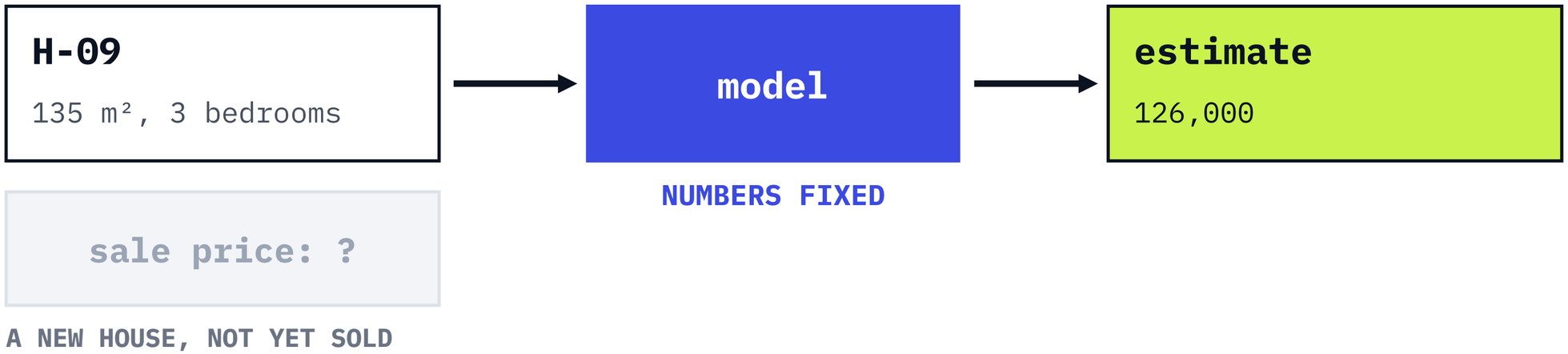
Suggested time · 8 minutes

Training: compare with the known answer



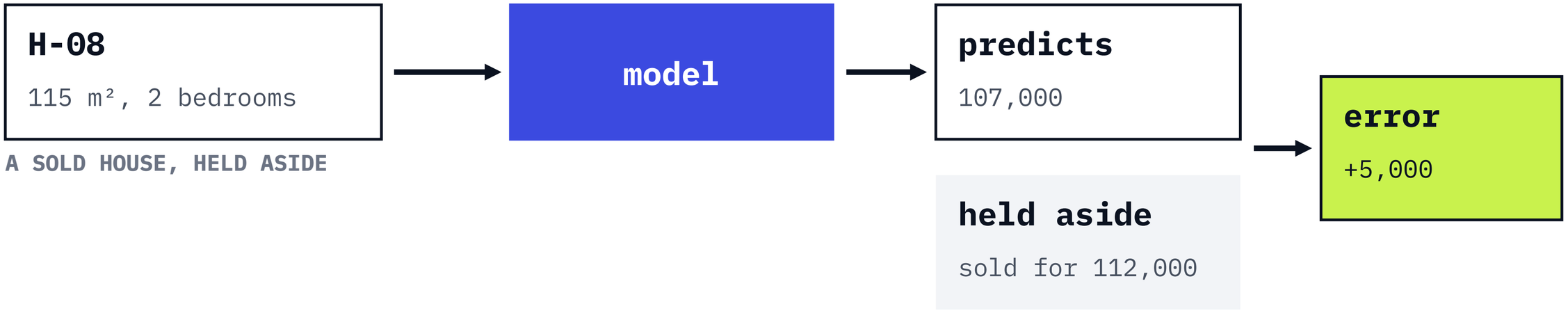
Training uses the errors on all the training examples to set the model's numbers. A straight line can also be solved in one calculation.

Inference: a new house, features only



Only the features go in. Making a prediction does not retrain the model.

Checking: an answer held aside



Keep some examples with known answers away from training, predict them, and compare. That is how we judge a model.

INSTRUCTOR DO

Step 4.7

From one feature to six

The words on the salary table, then 1,338 insured people.

Suggested time · 8 minutes

THE SIX WORDS, ON THE SALARY TABLE

Each word, in the model you trained

Word	In the salary model
Example	one past hire
Feature	years_experience
Label	salary
Identifier	none in this table
Model	two numbers: 25,792 and 9,450
Prediction	about 73,042 for 5 years

A SECOND DATASET · 1,338 INSURED PEOPLE

Six features, one label

FEATURES

LABEL

age	sex	bmi	children	smoker	region	charges
19	female	27.9	0	yes	southwest	**16,885**
18	male	33.8	1	no	southeast	**1,726**
28	male	33.0	3	no	southeast	**4,449**
33	male	22.7	0	no	northwest	**21,984**

One row is one insured person. charges: yearly medical charges in US dollars. The file says yes or no; Auto detect loads smoker as BOOLEAN, true or false.

DEFINITION

One-hot encoding: each text value becomes its own column of 0s and 1s.

One text column: region

Eight people from the insurance data. **region** holds words, and a model can only multiply numbers.

region
southwest
southeast
northwest
northeast
southeast
northwest
northeast
southwest

The same column, as four columns of 0s and 1s

region	northeast	northwest	southeast	southwest
southwest	0	0	0	**1**
southeast	0	0	**1**	0
northwest	0	**1**	0	0
northeast	**1**	0	0	0
southeast	0	0	**1**	0
northwest	0	**1**	0	0
northeast	**1**	0	0	0
southwest	0	0	0	**1**

Each row has exactly one 1. BigQuery does this for every text and true/false feature when it trains: region, sex, smoker.

Drop one column; it becomes the baseline

region	northwest	southeast	southwest
southwest	0	0	**1**
southeast	0	**1**	0
northwest	**1**	0	0
northeast	0	0	0
southeast	0	**1**	0
northwest	**1**	0	0
northeast	0	0	0
southwest	0	0	**1**

Three columns for four regions. A row of all 0s means northeast, the baseline.

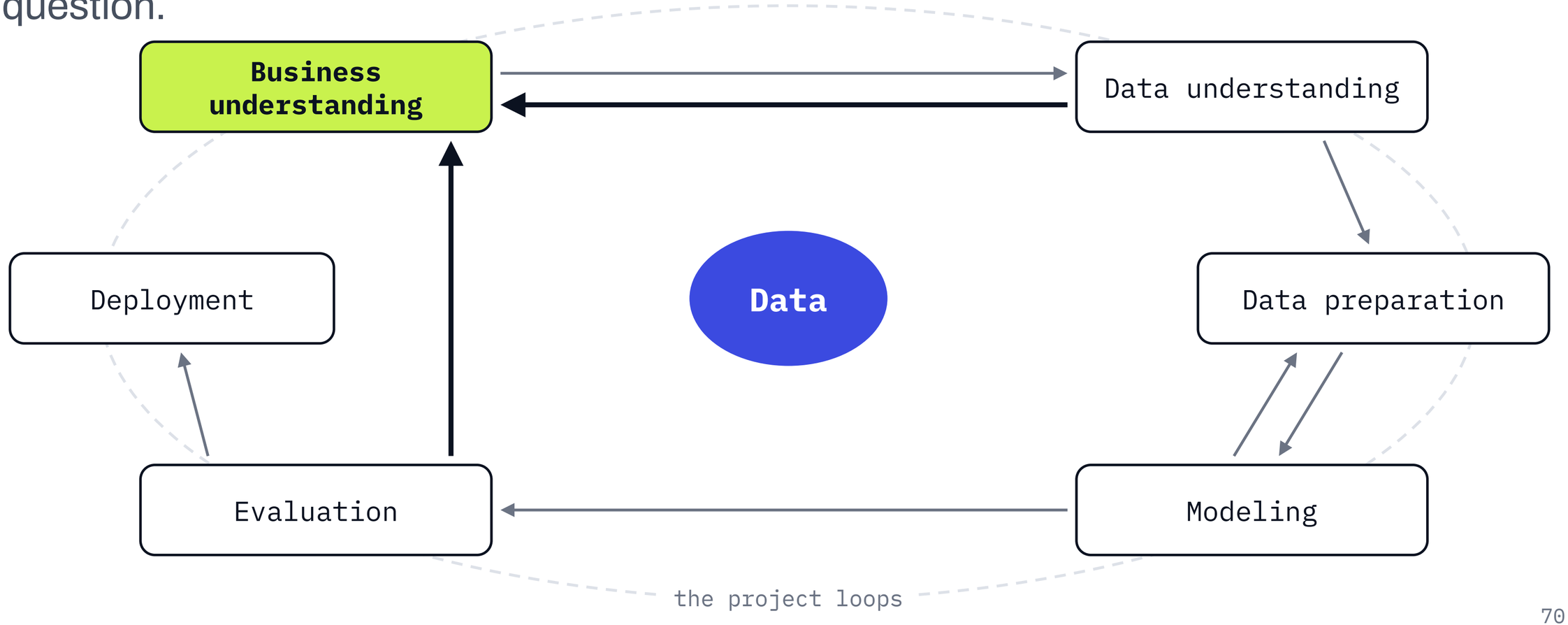
CATEGORY_ENCODING_METHOD

One-hot or dummy?

	ONE_HOT_ENCODING (default)	DUMMY_ENCODING
Columns for 4 regions	4	3: one value is the baseline
A weight means	that region’s own effect	the difference from the baseline
Use it when	you want every value’s weight	you want p-values, or "compared with …"
Predictions	the same	the same

Six phases, and the arrows go back

Two pairs of phases go back and forth; evaluation can send you back to the business question.



INSTRUCTOR DO

Step 4.8

Held aside, and why

A score on rows the model has seen flatters it.

Suggested time · 8 minutes

It looks excellent

ML.EVALUATE scores a model. MAE: the average miss. R^2: 1 is perfect.

```
SELECT ROUND(mean_absolute_error) AS mae,
       ROUND(r2_score, 3) AS r2
FROM ML.EVALUATE(MODEL ml.salary_model);
```

```
→ 1 row · MAE ≈ 4,644 · R² ≈ 0.957
```

BIGQUERY'S DEFAULT SPLIT, AUTO_SPLIT

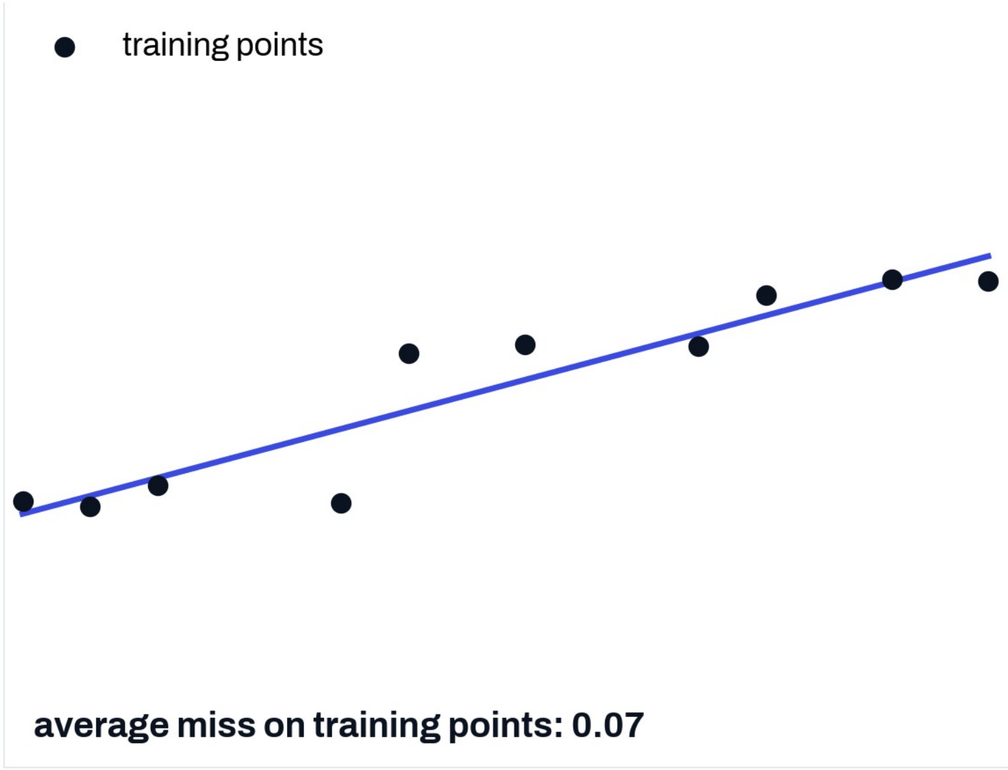
Which rows was that measured on?

Rows in the training data	What BigQuery holds aside to score
fewer than 500	**nothing: scored on training rows**
500 to 50,000	20% of the rows
more than 50,000	10,000 rows

The salary table has 30 rows. Its 0.957 was measured on the 30 hires it learned from.

Learn the pattern, or memorise the points

A straight line



average miss on training points: 0.07

A wiggly curve (degree 9)

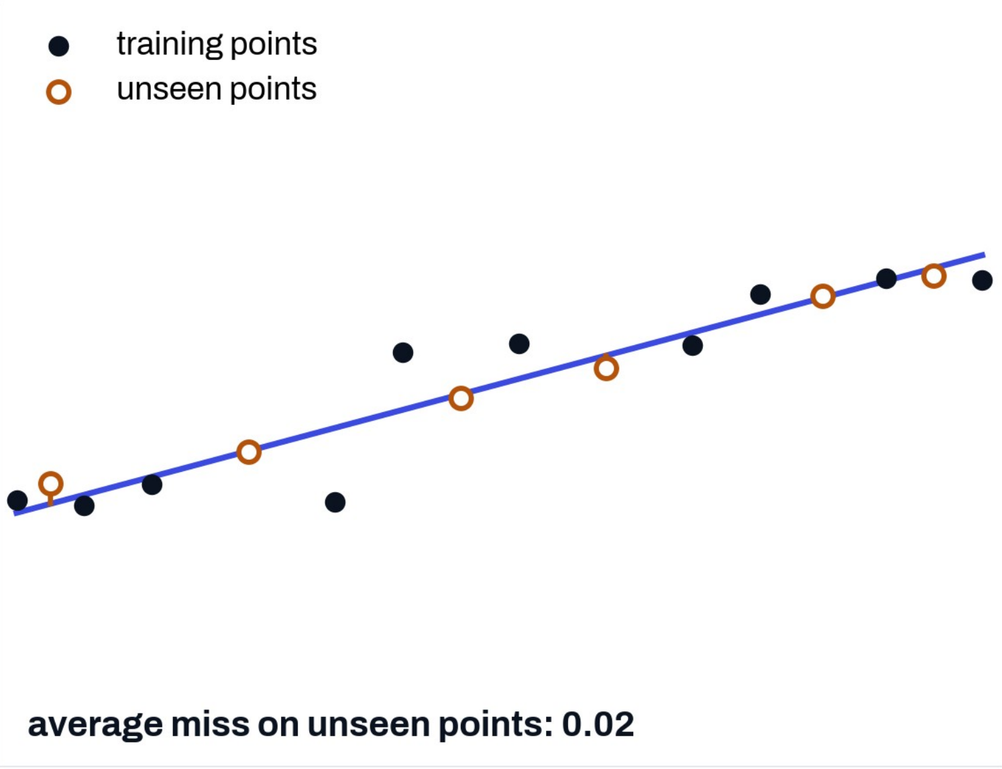


average miss on training points: 0.00

Invented points. The curve is a degree-9 polynomial: it passes through every training point.

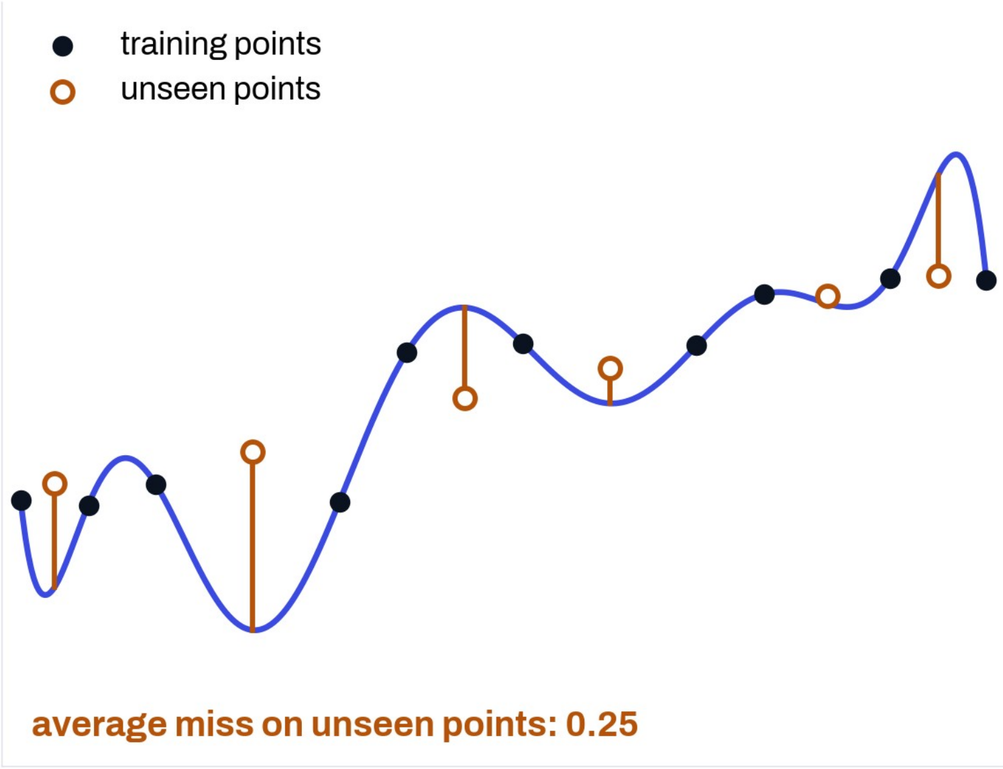
Perfect on training, poor on new points

A straight line



average miss on unseen points: 0.02

A wiggly curve (degree 9)



average miss on unseen points: 0.25

Overfitting: fitting the training rows so closely that the model does worse on rows it has not seen.

THE FIX, IN TWO OPTIONS

Hold 20% aside, on purpose

`data_split_method = 'RANDOM'` with an eval fraction of 0.2.

```
CREATE OR REPLACE MODEL ml.insurance_model
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['charges'],
  data_split_method = 'RANDOM',
  data_split_eval_fraction = 0.2
) AS
…
```

DATA_SPLIT_METHOD · FIVE VALUES · LAB 05

Five ways to choose the rows held aside

Method	What is held aside	Use it when
AUTO_SPLIT	none under 500 rows; otherwise 20%, or 10,000 rows	the default: check what it chose
RANDOM	a random share: DATA_SPLIT_EVAL_FRACTION	the rows have no order
CUSTOM	**the rows where your TRUE/FALSE column is TRUE**	**you choose, or several models must share rows**
SEQ	the last rows, sorted by a column	time order: learn the past, score the latest
NO_SPLIT	nothing	you made the groups yourself

DATA_SPLIT_TEST_FRACTION adds a third, test group, but only when BigQuery tunes a model's settings for you.

EVERYONE DO

Step 4.9

Five questions

One line each in the chat.

Suggested time · 3 minutes

CHECK BEFORE WE SCORE

Five questions, one line each

#	Question
1	In ml.salary_data: one row is ___? The feature? The label?
2	Which statement trained the model? Which made a prediction?
3	Regression or classification today? Why?
4	What did ML.WEIGHTS show?
5	Why hold rows aside before scoring?

Five answers

#	Answer
1	One past hire; years_experience; salary
2	CREATE MODEL trained; ML.PREDICT predicted
3	Regression: the label is a number
4	The two numbers the model learned: 25,792 and 9,450
5	A score on rows the model has seen flatters it

PART 04

Many features, scored honestly

1,338 insured people, 20% held aside, and a guess the model must beat.

EVERYONE DO

Step 5.1

Upload and look at the insurance data

One row per insured person, six features.

Suggested time · 8 minutes

labs/D11/03-Evr-Many-Features/README.md

What will a customer’s yearly medical charges be? Six facts about each person to learn from.

Smokers against non-smokers

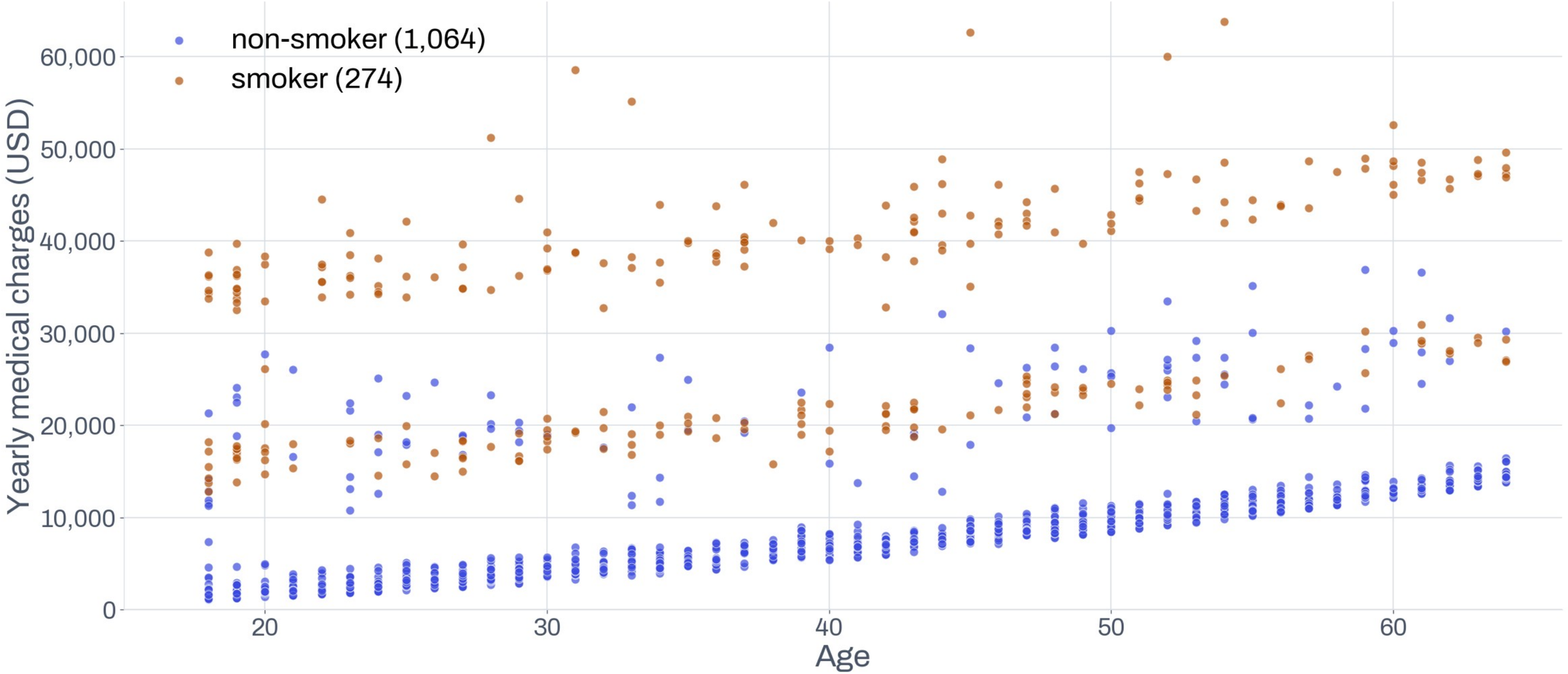
GROUP BY smoker: SQL you know, before any model.

```
SELECT smoker,
       COUNT(*) AS people,
       ROUND(AVG(charges)) AS avg_charges
FROM ml.insurance
GROUP BY smoker;
```

→ 2 rows · true 274, ≈ 32,050 · false 1,064, ≈ 8,434

ALL 1,338 PEOPLE

Age against charges



Medical Cost Personal Datasets (Kaggle), from Lantz, Machine Learning with R. US dollars.

EVERYONE DO

Step 5.2

Train with six features, holding 20% aside

About a minute. Then five people.

Suggested time · 10 minutes

labs/D11/03-Evr-Many-Features/README.md

BLOCK 2 · TRAIN

Six features, 20% held aside

data_split_method: about 268 people kept aside for scoring.

```
CREATE OR REPLACE MODEL ml.insurance_model
OPTIONS (
  model_type = 'LINEAR_REG',
  input_label_cols = ['charges'],
  data_split_method = 'RANDOM',
  data_split_eval_fraction = 0.2
) AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance;
```

Five people: actual against predicted

ML.PREDICT adds predicted_charges; the other columns pass through.

```
SELECT age, smoker, bmi,
       ROUND(charges) AS charges,
       ROUND(predicted_charges) AS predicted
FROM ML.PREDICT(MODEL ml.insurance_model,
  (SELECT * FROM ml.insurance))
LIMIT 5;
```

→ 5 rows · most close; a few far off

INSTRUCTOR DO

Step 5.3

MAE by hand

Five past hires and the salary model’s own two numbers.

Suggested time · 8 minutes

Five hires, five errors

Years	Salary	25,792 + 9,450 × years	Error
1.1	39,343	36,187	**+3,156**
1.3	46,205	38,077	**+8,128**
1.5	37,731	39,967	**−2,236**
2.0	43,525	44,692	**−1,167**
2.2	39,891	46,582	**−6,691**

The first five rows of salary_data.csv; predictions from the model’s two numbers.

MEAN ABSOLUTE ERROR

MAE: the average miss, without signs

(3,156 + 8,128 + 2,236 + 1,167 + 6,691) ÷ 5

MAE ON THESE FIVE

4,276

dollars

For a manager: "on these five hires, the line misses by about 4,300 dollars on average."

EVERYONE DO

Step 5.4

Score it, against the average guess

The 20% it never saw, then a guess that uses no features.

Suggested time · 14 minutes

labs/D11/03-Evr-Many-Features/README.md

BLOCK 4 · SCORE ON THE 20%

The score on people it never saw

ML.EVALUATE with no table: the rows held aside.

```
SELECT ROUND(mean_absolute_error) AS mae,
       ROUND(SQRT(mean_squared_error)) AS rmse,
       ROUND(r2_score, 3) AS r2
FROM ML.EVALUATE(MODEL ml.insurance_model);
```

→ MAE ≈ 4,200 · RMSE ≈ 6,100 · R² ≈ 0.75

Your split is random, so your three numbers move a little from these.

The average, for everyone

AVG(ABS(…)): the MAE of a guess that uses no features.

```
SELECT ROUND(AVG(ABS(charges - 13270))) AS mae_of_the_guess
FROM ml.insurance;
```

→ 1 row · ≈ 9,091

THE MODEL AGAINST THE GUESS

Less than half the miss

GUESS 13,270 FOR EVERYONE

≈ 9,091

its average miss (MAE), dollars

THE MODEL

≈ 4,200

its average miss (MAE), dollars

Both are misses. 13,270 is the guess itself; 9,091 is how far that guess lands from people’s real charges, on average.

ONE HELD-OUT 20% · TYPICAL VALUES

Three ways to score the misses

Score	What it says	Units	The model	Guess the average
MAE	the average miss	dollars	**≈ 4,200**	≈ 9,100
RMSE	the average miss, big misses counting more	dollars	**≈ 6,100**	≈ 12,100
R^2	the share of the variation explained	none, 0 to 1	**≈ 0.75**	0

RMSE is above MAE because a few people are missed by a lot: about 1 in 11 by more than 10,000 dollars.

DEFINITION

R^2: the share of the variation in the label that the model explains, compared with guessing the average for everyone. 0 is no better than the guess; 1 is perfect.

R² FROM THE TWO RMSES

Where 0.75 comes from

$$R^2 = 1 - (\text{model's RMSE} \div \text{the guess's RMSE})^2$$

$$= 1 - (6{,}100 \div 12{,}100)^2 \approx 1 - 0.25 = 0.75$$

The model explains about three-quarters of why charges differ from person to person. A quarter is left: things these six columns do not know.

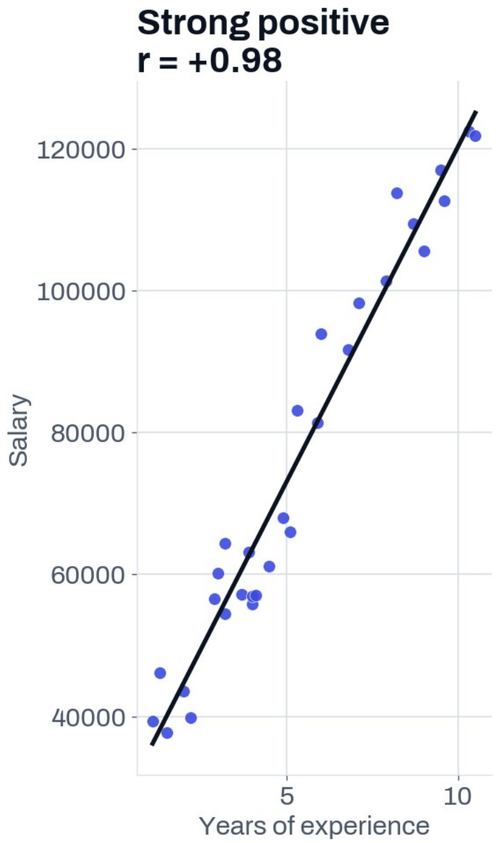
READING R^2

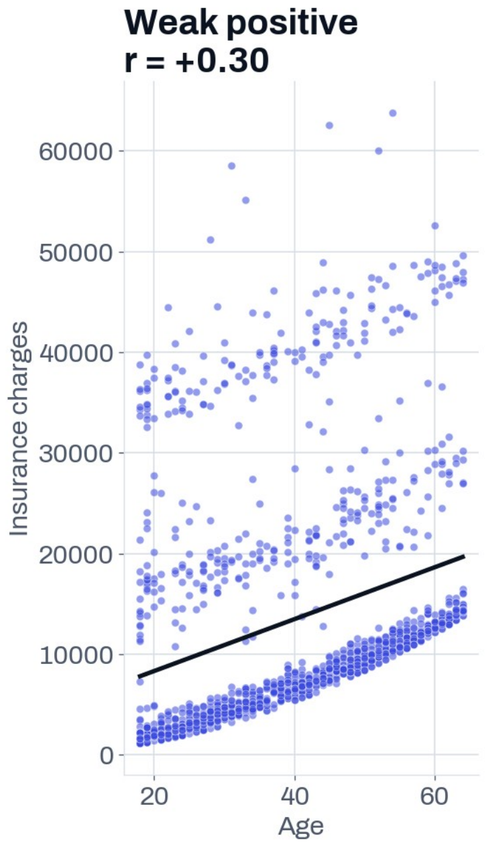
What an R^2 says

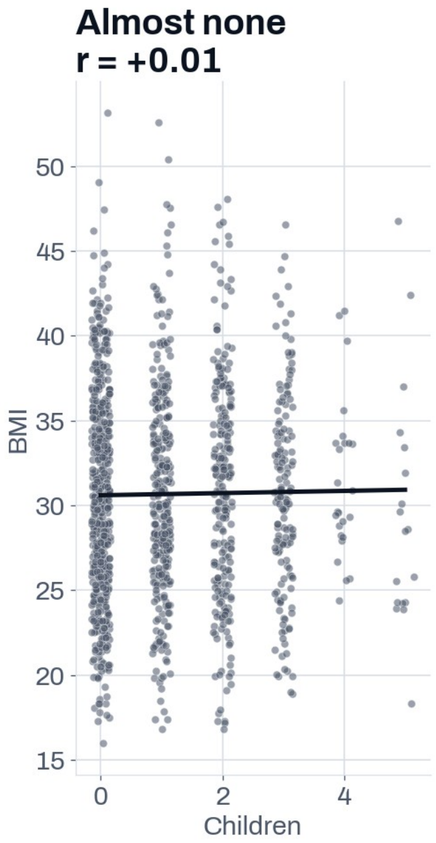
R^2	What it means
1	perfect: on unseen rows, suspect a leak, not a great model
≈ 0.75 (ours)	**three-quarters of the variation explained; strong for data about people**
0	no better than guessing the average for everyone
below 0	worse than the guess: possible on unseen rows

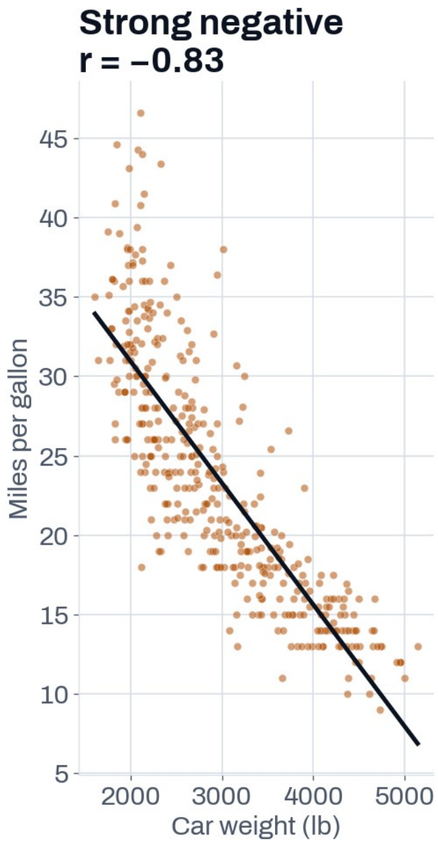
What counts as good depends on the field: models of human behaviour often stay below 0.5; physical measurements often pass 0.95.

Do two columns move together?









r runs from −1 to +1. With one feature, $R^2 = r^2$: $0.978^2 \approx 0.957$ for salary. In SQL: CORR(x, y).

MAE AND R^2, TOGETHER

Two questions, two numbers

	MAE	R^2
Question	how far off, in dollars?	how much of the variation is explained?
Ours	≈ 4,200 dollars: about a third of the average charge (13,270)	≈ 0.75
On its own	needs the scale: 4,200 off is small for a house, huge for a phone bill	reads the same whatever the units
Tell the business	"a typical quote misses by about 4,200"	"we explain three-quarters of why charges differ"

EVERYONE DO

Step 5.5

Seen rows against unseen rows

The check for overfitting, in one query.

Suggested time · 8 minutes

labs/D11/03-Evr-Many-Features/README.md

Two scores, side by side

UNION ALL: the held-aside score, then all rows (mostly seen).

```
SELECT 'held aside' AS scored_on, ROUND(r2_score, 3) AS r2
FROM ML.EVALUATE(MODEL ml.insurance_model)
UNION ALL
SELECT 'all rows', ROUND(r2_score, 3)
FROM ML.EVALUATE(MODEL ml.insurance_model, (SELECT * FROM ml.insurance));
```

→ 2 rows · both $R^2 \approx 0.75$

Break

30 minutes

After the break: what the model learned, two new people, and whether we need all six features.

PART 05

Read it, use it, simplify it

What it learned, two new people, fewer features, and the options to try next.

EVERYONE DO

Step 6.1

Read what it learned

One weight per feature, in dollars.

Suggested time · 8 minutes

labs/D11/03-Evr-Many-Features/README.md

Six features, six kinds of weight

category_weights: one weight per value of a text or true/false feature.

```
SELECT processed_input, ROUND(weight, 1) AS weight, category_weights
FROM ML.WEIGHTS(MODEL ml.insurance_model);
```

WHAT EACH WEIGHT SAYS

Read the weights as sentences

Feature	Weight	In words
age	≈ 257	about 257 dollars more for each year of age
bmi	≈ 339	about 339 more for each BMI point
children	≈ 476	about 476 more for each child
smoker	**true ≈ 23,800 above false**	**the same person, as a smoker: about 23,800 more**

EVERYONE DO

Step 6.2

Ask it about two new people

The same person, smoking or not.

Suggested time · 7 minutes

labs/D11/03-Evr-Many-Features/README.md

The same person, smoking or not

UNION ALL builds two rows the model has never seen. smoker is BOOLEAN: write **FALSE** and **TRUE**, not 'no' and 'yes'.

```
SELECT smoker, ROUND(predicted_charges) AS predicted_charges
FROM ML.PREDICT(MODEL ml.insurance_model,
  (SELECT 40 AS age, 'female' AS sex, 28.0 AS bmi, 1 AS children,
          FALSE AS smoker, 'northwest' AS region
   UNION ALL
   SELECT 40, 'female', 28.0, 1, TRUE, 'northwest'));
```

→ 2 rows · false ≈ 7,960 · true ≈ 31,810

STUDENT DO

Do we need all six?

Train a three-feature and a one-feature model. Score all three on the rows they held aside.

Suggested time

12 minutes

`labs/D11/04-Stu-Fewer-Features/README.md`

STUDENT DO

Three models, one comparison

INSTRUCTIONS	Block 1: age, bmi and smoker only. Block 2: age only. Block 3 scores all three together. Which would you give the pricing team? If two score within about 100 dollars, keep the simpler.
HINT	Each model holds aside its own random 20%: small differences are noise.
BONUS	Train on smoker alone. What does its R^2 tell the pricing team?

Full steps and Check yourself: labs/D11/04-Stu-Fewer-Features/README.md

REVIEW

Time's up. Let's review.

Do we need all six?

`Solved: labs/D11/04-Stu-Fewer-Features/Solved/fewer_features.sql`

THE ANSWER

Three features are enough

Model	Features	MAE	R^2
insurance_model	6	≈ 4,200	≈ 0.75
insurance_three	3	≈ 4,200	≈ 0.75
insurance_age	1	≈ 9,100	≈ 0.08

sex, children and region add almost nothing. Age alone cannot see the smokers, and smoking is the largest effect in the data.

INSTRUCTOR DO

Step 6.5

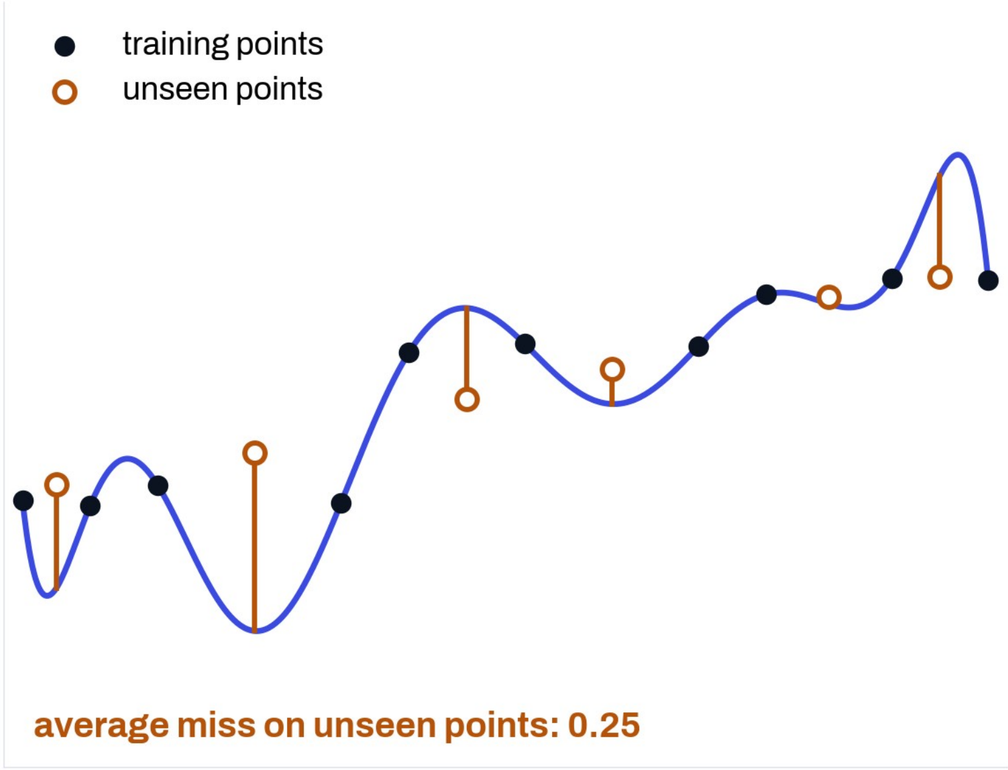
Regularization, and the other options

A cure for overfitting, then the CREATE MODEL page and the three labs.

Suggested time · 8 minutes

A penalty on large weights calms the curve

Degree 9, no penalty



Degree 9, with an L2 penalty



The same invented points as the overfitting slides. Right: the same degree-9 model, trained with an L2 penalty.

DEFINITION

Regularization: training aims for small errors plus small weights, so the model cannot chase every training point.

Ridge, Lasso and Elastic Net

	Ridge	Lasso	Elastic Net
In BigQuery	l2_reg > 0	l1_reg > 0	both > 0
Penalises	squared weights	weight sizes	both
The weights	all shrink; none reach 0	weak ones become 0	weak ones 0; the rest shared
Good for	features that overlap	keeping only useful features	many features, some overlapping

Ordinary linear regression is both at 0: BigQuery’s default. Next: all four fitted to the cars.

Elastic Net: turn both penalties on

No **ELASTIC_NET** model type: LINEAR_REG with l1_reg and l2_reg both above 0.

```
CREATE OR REPLACE MODEL ml.cars_enet
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['mpg'],
         data_split_method = 'RANDOM',
         data_split_eval_fraction = 0.2,
         l1_reg = 500, l2_reg = 100) AS
SELECT * FROM ml.cars_clean;
```

Strengths are on BigQuery’s own scale: lab 07 tries several.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

Ordinary regression: the equation to compare against

ORDINARY · NO PENALTY

```
mpg = 23.5
  − 0.61 × cylinders
  − 0.15 × displacement
  − 1.49 × horsepower
  − 4.54 × weight
  − 0.02 × acceleration
```

Weight does most of the work.

23.5 is the average mpg.

4.54 × weight: one typical spread heavier (847 lb), 4.5 mpg less.

Standardized weights (lab 06): one yardstick for every feature.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

Ridge: every weight held back, the credit shared

ORDINARY · NO PENALTY

```
mpg = 23.5
  − 0.61 × cylinders
  − 0.15 × displacement
  − 1.49 × horsepower
  − 4.54 × weight
  − 0.02 × acceleration
```

RIDGE · L2

```
mpg = 23.5
  − 1.15 × cylinders     larger
  − 1.34 × displacement  larger
  − 1.47 × horsepower
  − 2.36 × weight        smaller
  − 0.02 × acceleration
```

Weight’s −4.54 falls to −2.36; cylinders and displacement take up the slack. Sum of squared weights: 23.3 → 10.9. No weight reaches 0.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

Lasso: a weight of 0 drops the feature

ORDINARY · NO PENALTY

```
mpg = 23.5
  − 0.61 × cylinders
  − 0.15 × displacement
  − 1.49 × horsepower
  − 4.54 × weight
  − 0.02 × acceleration
```

LASSO · L1

```
mpg = 23.5
  − 0    × cylinders    dropped
  − 0.42 × displacement
  − 0.72 × horsepower
  − 3.98 × weight
  − 0    × acceleration dropped
```

0 × cylinders is 0 for every car: cylinders no longer moves the prediction. Displacement says much the same, so Lasso kept one and dropped the other.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

Elastic Net: drop the weakest, share the rest

ORDINARY · NO PENALTY

```
mpg = 23.5
  − 0.61 × cylinders
  − 0.15 × displacement
  − 1.49 × horsepower
  − 4.54 × weight
  − 0.02 × acceleration
```

ELASTIC NET · L1 + L2

```
mpg = 23.5
  − 1.08 × cylinders
  − 1.30 × displacement
  − 1.38 × horsepower
  − 2.30 × weight
  − 0    × acceleration dropped
```

Like Lasso, acceleration goes to 0. Like Ridge, the four engine features share the credit instead of one being dropped.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

Four fits of the same cars

Standardized weight	Ordinary	Ridge	Lasso	Elastic Net
cylinders	−0.61	−1.15	**0**	−1.08
displacement	−0.15	−1.34	−0.42	−1.30
horsepower	−1.49	−1.47	−0.72	−1.38
weight	−4.54	−2.36	−3.98	−2.30
acceleration	−0.02	−0.02	**0**	**0**
weights at 0	**0**	**0**	**2**	**1**

Every fit starts at 23.5, the average mpg. Only the weights change.

ILLUSTRATIVE · ALL 398 CARS · FITTED IN PYTHON

One car, four equations, four close answers

A car from the table: 4 cylinders, 116 cubic inches, 90 hp, 2,123 lb. It does 28 mpg.

Fit	Predicted mpg	Miss
Ordinary	29.3	1.3
Ridge	28.4	0.4
Lasso	28.1	0.1
Elastic Net	28.2	0.2

Different weights, nearly the same prediction: the features overlap, so many weightings fit.

ILLUSTRATIVE · AVERAGE OF 300 RANDOM 80/20 SPLITS

Smaller weights are not a better score

Fit	MAE on rows held aside	R^2 on rows held aside
Ordinary	**3.29**	**0.69**
Ridge	3.31	0.68
Lasso	3.48	0.65
Elastic Net	3.32	0.68

Five features and 398 cars: no penalty scores best here. Penalties pay off with many features and few rows, and too strong a penalty underfits. Choose with ML.EVALUATE.

Lasso in BigQuery: a much bigger number

Strengths in the tens kept every weight. **`l1_reg = 500`** put cylinders at 0.

```
CREATE OR REPLACE MODEL ml.cars_lasso
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['mpg'],
         data_split_method = 'RANDOM',
         data_split_eval_fraction = 0.2,
         optimize_strategy = 'BATCH_GRADIENT_DESCENT',
         l1_reg = 500, early_stop = FALSE) AS
SELECT * FROM ml.cars_clean;
```

Lab 07, step 4: find the strength that drops a feature in your split.

OPTIMIZE_STRATEGY · GOOGLE’S RULES

How BigQuery trains each method

Method	AUTO_STRATEGY trains it by	NORMAL_EQUATION allowed?
Ordinary, Ridge	one calculation, on small data like ours	yes
Lasso, Elastic Net	**gradient descent, in steps**	**no: l1_reg rules it out**

early_stop = FALSE lets every step run, so the penalty shows. Check the result: weight = 0 in ML.WEIGHTS, one row per step in ML.TRAINING_INFO.

CREATE MODEL · LINEAR REGRESSION OPTIONS

Change one at a time, watch what happens

Options	What they change	Lab
DATA_SPLIT_METHOD, DATA_SPLIT_EVAL_FRACTION	which rows are held aside	05
FIT_INTERCEPT	whether the line has a starting point	06
CATEGORY_ENCODING_METHOD	how text becomes numbers	06
L1_REG, L2_REG	how hard large weights are held back	07
OPTIMIZE_STRATEGY, MAX_ITERATIONS	how training runs	07

docs.cloud.google.com/bigquery/docs/reference/standard-sql/bigqueryml-syntax-create-glm

PARTNER DO

Would you price with it?

Average miss about 4,200 dollars: about 8,600 for smokers, 3,000 for non-smokers. Good enough?

Suggested time

10 minutes

`labs/D11/03-Evr-Many-Features/README.md`

PARTNER DO

Answer the insurer in three sentences

INSTRUCTIONS	The insurer wants to set next year’s price per customer with this model. Is an average miss of about 4,200 dollars good enough? For which customers is it not? Should sex be a feature in a price at all?
HINT	Use the weights and the two scores.
BONUS	What would you ask for before you signed off on it?

Full steps and Check yourself: labs/D11/03-Evr-Many-Features/README.md

STUDY LIST · IN CLASS AND IN THE LABS

BigQuery ML functions today

In class	What it gives you	Activity
ML.PREDICT	a prediction for each row you pass in	02, 03
ML.EVALUATE	the scores on the rows you pass: MAE, R^2 and more	02–04
ML.WEIGHTS	the numbers the model learned, feature by feature	02, 03

New in the labs	What it gives you	Lab
ML.TRAINING_INFO	each training step and its loss	07
ML.ADVANCED_WEIGHTS	the weights with p-values (bonus)	06

EVERY MODULE 3 FUNCTION · resources/bigquery-ml-functions.md

CLOSE

What you can do now

- Place ML in AI; name features and label
- Train, read and use LINEAR_REG
- Score on held-out rows against a guess
- Keep the simpler model when scores match

NEXT · D12

“Will this employee leave?” A1 due Wednesday.

LAB

AFTER CLASS · SELF-GUIDED

The options behind CREATE MODEL

TASK

Three short labs, one option at a time: splits; intercept and text; regularization and training.

FILES

`labs/D11/05-Lab-Data-Splits/README.md`

`labs/D11/06-Lab-Intercept-And-Categories/README.md`

`labs/D11/07-Lab-Regularization-And-Training/README.md`

DONE WHEN

Each lab ends with a table of what you saw.

Optional self-study: lab 08, Nakheel end to end.